

Question 1

Two particles are moving uniformly in two-dimensional space on straight lines. The first particle starts at point $\vec{r}_1 = (0, 4) \text{ m}$ with velocity $\vec{v}_1 = (1, -2) \text{ m/s}$, and the second particle starts at point $\vec{r}_2 = (-3, 2) \text{ m}$ with velocity $\vec{v}_2 = (2, 1) \text{ m/s}$. Find:

- The vector of their relative position with respect to time, $\vec{R}(t)$. — 0.5 points
- The moment t_c of closest approach and its distance. — 0.3 points
- The intersection point of their trajectories. — 0.2 points

Solution:

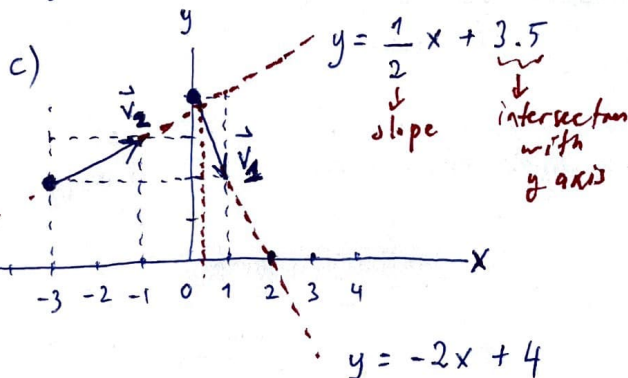
$$a) \vec{r}_1(t) = \vec{r}_1 + \vec{v}_1 t = (t, 4 - 2t) \text{ m}$$

$$\vec{r}_2(t) = \vec{r}_2 + \vec{v}_2 t = (-3 + 2t, 2 + t) \text{ m}$$

$$\vec{R}(t) = \vec{r}_2(t) - \vec{r}_1(t) = (-3 + t, -2 + 3t) \text{ m}$$

$$b) R^2(t) = (t-3)^2 + (3t-2)^2 = 10t^2 - 18t + 13$$

$$\frac{d}{dt} [R^2(t)] = 20t - 18 = 0 \Rightarrow t_c = \frac{9}{10} \text{ s}$$



$$y = \frac{1}{2}x + 3.5 \Rightarrow -2x + 4 = \frac{1}{2}x + 3.5$$

$$y = -2x + 4 \Rightarrow x = \frac{1}{5}$$

$$\Rightarrow y = \frac{18}{5}$$

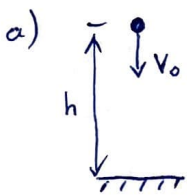
$$(x, y) = \left(\frac{1}{5}, \frac{18}{5} \right) \text{ intersection point.}$$

Question 2

An object is thrown vertically downward from a height h with an initial velocity v_0 . The motion takes place on earth with gravitational acceleration g and air resistance is neglected.

- (a) What should be v_0 so that the object lands τ seconds earlier than it would if it was released from rest? — 0.7 points
- (b) Assume that there was also a force acting in the perpendicular direction as the throw of the ball and accelerating it as $a = \frac{g}{\sqrt{3}}$. What would be the range of the object in this case? — 0.3 points

Solution:



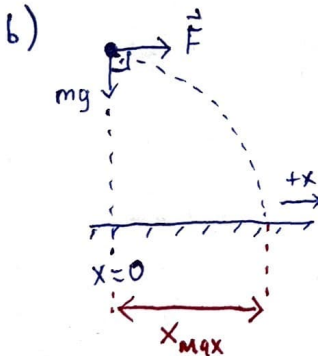
If it was released from rest:

$$h = \frac{1}{2} g t_{\text{rest}}^2 \rightarrow t_{\text{rest}} = \sqrt{\frac{2h}{g}}$$

But we want:

$$h = v_0 (t_{\text{rest}} - \tau) + \frac{1}{2} g (t_{\text{rest}} - \tau)^2$$

$$\Rightarrow v_0 = \frac{h}{t_{\text{rest}} - \tau} - \frac{1}{2} g (t_{\text{rest}} - \tau) = \boxed{\frac{h}{\sqrt{\frac{2h}{g}} - \tau} - \frac{1}{2} g \left(\sqrt{\frac{2h}{g}} - \tau \right)}$$



The motion in vertical axis will not change; it will still take $t_{\text{rest}} - \tau$ seconds for the particle to fall down.

In x -direction:

Instead of falling down at point $x=0$, the object will cover a distance (its range) in x -direction.

$$x(t) = x_0 + v_x(t) t + \frac{1}{2} \frac{g}{\sqrt{3}} t^2$$

$$x_{\text{max}} = x(t_{\text{rest}} - \tau) = \frac{1}{2} \frac{g}{\sqrt{3}} (t_{\text{rest}} - \tau)^2 = \boxed{\frac{1}{2} \frac{g}{\sqrt{3}} \left(\sqrt{\frac{2h}{g}} - \tau \right)^2}$$

Question 3

A particle of mass m can move without friction along the x -axis. At time $t = 0$, it is at rest at position $x = 0$. A force $F_x(t) = a + bt$ (where a and b are constants) acts on it.

(a) Determine the acceleration $a(t)$, the velocity $v(t)$, the displacement $x(t)$. — 0.6 points

(b) Determine the power delivered by the force at time t . — 0.4 points

Solution:

a) $a(t) = m(a + bt) = ma + mbt$

$$v(t) = \int ma dt + \int mbt dt + C = mat + mb \frac{t^2}{2} + C$$

$$v(0) = C = 0 \text{ (the particle was at rest at } t=0) \Rightarrow v(t) = mat + mb \frac{t^2}{2}$$

$$x(t) = \int mat dt + \int mb \frac{t^2}{2} dt + D = m \frac{at^2}{2} + mb \frac{t^3}{6} + D$$

$$x(0) = D = 0 \text{ (the particle was at the position } x=0 \text{ at time } t=0)$$

$$\Rightarrow x(t) = m \frac{at^2}{2} + mb \frac{t^3}{6}$$

b) Work done on the particle as it moves from $x=0$ to x ;

$$W = \int_0^x F_x dx = \int_0^x (a + bt) dx = \int_0^x a dx + \int_0^x bt dx = ax + b \int_0^x t \frac{dx}{dt} dt$$

$$= ax + b \int_0^t t v(t) dt = ax + b \int_0^t [mat^2 + mb \frac{t^3}{2}] dt$$

$$= ax + b \left(m \frac{at^3}{3} + m b \frac{t^4}{8} \right) = m \frac{a^2 t^2}{2} + b m a \frac{t^3}{6} + \frac{2 \cdot b m a t^3}{2 \cdot 3} + m b^2 \frac{t^4}{8}$$

$$W(t) = \left(\frac{ma^2}{2} \right) t^2 + \left(\frac{mba}{2} \right) t^3 + \left(\frac{mb^2}{8} \right) t^4$$

$$P(t) = \frac{W(t)}{t} = \left(\frac{ma^2}{2} \right) t + \left(\frac{mba}{2} \right) t^2 + \left(\frac{mb^2}{8} \right) t^3$$

at point x , time is t

the velocity of the particle at time t and position x

$V(t)$

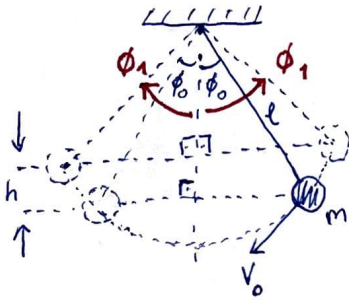
When $x=0$, $t=0$

Question 4

A simple pendulum with mass m and length l , initially displaced by an angle ϕ_0 from the vertical and given an initial tangential speed v_0 at $t = 0$. Find the oscillation amplitude ϕ_1 .

— 1 point

Solution:



$$\frac{1}{2} m v_0^2 = m g h$$

$$h = l \cos \phi_0 - l \cos \phi_1 = l (\cos \phi_0 - \cos \phi_1)$$

$$\frac{1}{2} m v_0^2 = m g l (\cos \phi_0 - \cos \phi_1)$$

$$\frac{v_0^2}{2 g l} = \cos \phi_0 - \cos \phi_1$$

$$\cos \phi_1 = \cos \phi_0 - \frac{v_0^2}{2 g l}$$

$$\phi_1 = \arccos \left(\cos \phi_0 - \frac{v_0^2}{2 g l} \right)$$